PREDICTING SPEAKER PERFORMANCE IN DEBATES

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ABSTRACT

This paper studies individual and team performance in competitive debating using a real-world dataset from debate tournaments covering multiple seasons and thousands of speaker-level observations. Competitive debating provides a structured environment in which performance is repeatedly evaluated by external judges under standardized rules, making it a suitable setting for analyzing individual ability, experience, and peer interactions. The empirical analysis combines econometric and machine learning methods. Individual performance, measured by speaker points, is analyzed using pooled OLS and speaker fixed effects models to control for observable characteristics and unobserved individual heterogeneity, with dynamic specifications capturing learning and persistence. The results show no evidence of a gender gap in performance, while teammate quality emerges as a strong and robust determinant of individual scores, indicating substantial peer effects and sorting within teams. Experience also has a positive effect on performance. Complementary machine learning models based on gradient boosted trees are used to assess out-of-sample predictability. Speaker-level performance is moderately predictable, while team-level outcomes can be predicted with relatively high accuracy, despite weak explanatory power in simple linear team-level regressions. Text-based models are further applied to test for systematic motion-level bias but find no predictive signal.

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1 INTRODUCTION

Competitive debating provides a structured environment in which individual performance is repeatedly evaluated under varying strategic and social conditions. Speakers formulate arguments, respond to opponents, and coordinate with teammates, while being assessed by external judges using standardized scoring rules. This setting offers a natural context for studying how individual ability, experience, and peer interactions affect performance over time, with real participants facing meaningful incentives and reputational concerns.

This paper uses a unique real-world dataset from organized debate tournaments, covering multiple seasons and a large number of individual speaker appearances. The data include detailed information on speaker scores, team composition, debate motions, and tournament stages, allowing speakers to be followed across time and competitive contexts. The panel structure makes it possible to study both cross-sectional differences in performance and within-speaker dynamics such as learning and persistence.

The analysis combines econometric and machine learning methods. Individual performance is examined using pooled OLS and speaker fixed effects models, complemented by dynamic specifications to capture experience and persistence. In addition, gradient boosted tree models are used to assess the out-of-sample predictability of speaker and team outcomes, and text-based methods are applied to test for systematic motion-level bias.

2 DATA

The primary outcome variable is the number of points awarded in a debate, which reflects judges' assessments of individual performance. This outcome is observed alongside rich contextual information, including the side taken in the debate, the position of the speaker within the team, the debated motion, and the stage of the tournament at which the debate occurs. The availability of precise timing information enables the construction of dynamic measures such as experience and lagged performance, which are central for studying persistence and learning. By observing the same individuals across different team compositions and debate settings, the data allow for the separation of stable individual traits from situational and strategic influences on performance.

This dataset has been created in a way, that the further analysis can separate individual ability from peer and contextual effects. The data include information on teammates' performance, which is used to construct measures of average teammate quality and its lagged values. These variables capture the idea that speakers may learn from, be motivated by, or be constrained by the performance level of their co-speakers. At the same time, early career performance is used as a proxy for speaker's natural ability to make sure we do not mistake real skill with the effect peers have on a speaker. Tournament progression is captured through round indicators. Additional controls account for systematic advantages associated with debate motions and the side taken, which helps isolate speaker-level effects from structural biases in debate outcomes.

For the purpose of the motion-bias analysis, motion topics are constructed by transforming the original categorical topic annotations into a structured numeric representation. Each debate motion is originally associated with up to three topic labels, each accompanied by an intensity score. All unique, non-missing topic labels observed in the data are first identified and used to define a fixed set of topic columns. For each speaker–debate observation, the score corresponding to a given topic is then assigned to the matching topic column, while topics not associated with the motion are set to zero. This procedure allows a single motion to load on multiple topics with varying intensities, rather than forcing a mutually exclusive classification.

3 METHODS

3.1 Econometric framework

As a starting point, pooled ordinary least squares (OLS) model is estimated to provide a benchmark that ignores unobserved individual effects. While simple and interpretable, pooled OLS may be biased if unobserved speaker-specific traits, such as innate ability or motivation, are correlated with the explanatory variables. To address this concern, the analysis proceeds by estimating both fixed effects and random effects panel models. Fixed effects models control for time-invariant unobserved heterogeneity at the speaker level by relying exclusively on within-speaker variation, while random effects models exploit both within- and between-speaker variation under the assumption that individual effects are uncorrelated with the regressors. The suitability of the random effects specification is formally assessed using Breusch-Pagan Lagrange multiplier tests and Hausman tests comparing fixed and random effects estimates.

To further relax the strict exogeneity assumptions of the random effects model, we use a Mundlak-type specification. Collinearity diagnostics, rank checks, and condition number statistics are used to assess potential multicollinearity arising from the inclusion of mean terms. Inference in selected pooled specifications relies on heteroskedasticity-robust standard errors clustered at the speaker level, accounting for serial correlation and heteroskedasticity within individual panels.

Finally, the analysis extends to a dynamic panel data setting to explicitly model persistence in individual performance. A generalized method of moments (GMM) estimator is employed, in which current speaker performance depends on its own lag as well as on contemporaneous covariates capturing baseline ability, peer effects, and tournament progression. Lagged levels of the dependent variable are used as instruments to address endogeneity arising from dynamic feedback and unobserved individual effects. The model is estimated using a two-step difference GMM procedure with individual effects eliminated via first differencing. This dynamic specification allows the analysis to distinguish short-run peer and contextual effects from longer-run performance persistence.

3.2 Machine learning models

We apply supervised machine learning techniques to model motion-level affirmative-side advantage using only the textual content of debate motions. First, a continuous outcome variable is constructed at the motion–tournament level by aggregating debate outcomes by side. For each motion within a tournament, ballots gained by affirmative and negative teams are summed and normalized, and the difference between these aggregates defines the dependent variable, capturing systematic side advantage associated with a given motion. Each motion is tokenized into individual words and transformed into a numerical representation using term frequency–inverse document frequency weighting. This approach emphasizes terms that are distinctive to particular motions while downweighting commonly occurring words.

Model estimation is conducted using gradient boosted decision trees implemented via the `xgboost` framework, with the affirmative-side bias treated as a continuous regression target. To reflect the temporal structure of tournaments and avoid look-ahead bias, model training and evaluation follow a rolling-origin scheme based on tournament order: for each evaluation step, the model is trained on all tournaments observed up to that point and tested on the immediately subsequent tournament. This design approximates a real-time forecasting setting in which future motions are predicted using only past information. Model performance is assessed using standard out-of-sample metrics, including root mean squared error and the coefficient of determination relative to a constant-mean benchmark.

4 RESULTS AND DISCUSSION

4.1 Parametric Inference: Individual Speaker Performance

Table 1 reports the main results from pooled OLS and speaker fixed effects regressions explaining individual speaker performance, measured by awarded speaker points. There is no evidence of a systematic gender gap in performance. The coefficient on the male indicator is small in magnitude and statistically insignificant, indicating that male and female speakers receive comparable scores once observable characteristics are taken into account. This finding is robust to the inclusion of speaker fixed effects, suggesting that gender does not play a role either through between-speaker differences or within-speaker performance variation over time. Similarly, variables capturing tournament-specific conditions, such as the tournament round and motion balance, do not exhibit robust or statistically significant effects, implying that these contextual factors do not systematically influence individual scores once speaker-level characteristics are controlled for.

Table 1: Determinants of Individual Speaker Performance

	Dependent variable: Speaker points	
	(1) Pooled OLS	(2) Speaker FE
Male	-0.048	
	(0.241)	
Years since first debate	0.591***	
	(0.096)	
Tournament round	0.009	-0.036
	(0.092)	(0.079)
Motion balance × Affirmative	0.020	-0.055
	(0.149)	(0.129)
Motion balance	-0.358	0.095
	(0.392)	(0.347)
Average teammate score	0.746***	0.607***
	(0.030)	(0.032)
Constant	18.457***	
	(2.346)	
Observations	667	667
R-squared (overall)	0.586	0.394
Within R-squared (for FE)		0.394
Speaker fixed effects	No	Yes

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In contrast, the results highlight two economically and statistically important determinants of speaker performance. First, experience, which is proxied by years since the first recorded debate, has a strong and highly significant positive effect in the pooled specification, which means that speakers become better over time. Second, and most importantly, average teammate score enters positively and is highly significant in both models. This indicates that speakers tend to perform better when paired with stronger teammates, even after controlling for time-invariant individual ability. The persistence of this effect under speaker fixed effects suggests genuine peer or team effects rather than simple sorting on unobserved talent. Taken together, the findings imply that strong speakers tend to cluster in stronger teams, and that "natural talent," plays an important role in performance in later competitions.

Table 2: Econometric Model of Team-Level Debate Outcomes

	Ballots won per debate
Proportion male	-0.001
	(0.004)
Average team experience	-0.001
	(0.001)
Constant	0.521***
	(0.003)
Observations	506
R-squared	0.001

Notes: The dependent variable equals total ballots won divided by the number of debates and normalized by three judges. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2 presents results from a team-level regression explaining debate outcomes measured by ballots won per debate. Neither the proportion of male speakers on a team nor the average team experience exhibits a statistically significant effect on team success, with both coefficients close to zero in magnitude. The very low R^2 indicates that observable team composition characteristics explain little of the variation in ballots won across debates. These results suggest that, unlike individual speaker scores, aggregate team outcomes are not systematically driven by gender composition or accumulated experience, and are likely influenced by other elements.

4.2 Predictive Modeling: Team Outcomes and Motion Bias

Using a rolling out-of-sample validation design, where tournaments 307-$(t-1)$ are used for training and tournament t for testing (i.e. tournaments 307-316 for training and testing on tournament 317; tournaments 307-317 for training and testing on tournament 318, ...), we obtain the following performance metrics.

Table 3: Speaker-Level Prediction Performance (XGBoost)

Dataset (tournament ID)	RMSE	MAE	R^2	SD(actual)
Overall	4.015	3.129	0.331	4.911
Tournament 317	3.659	2.911	0.325	4.460
Tournament 318	4.485	3.560	0.253	5.190
Tournament 319	3.066	2.320	0.547	4.570
Tournament 320	3.467	2.630	0.421	4.560

At the speaker level, the model exhibits meaningful predictive power across all test tournaments. The overall root mean squared error (RMSE) of 4.015 is substantially lower than the standard deviation of the realized speaker points, indicating that the model improves considerably upon a naive mean benchmark. The positive out-of-sample R^2 of 0.331 confirms that a non-trivial share of variation in individual performance can be explained using the available features, even when predictions are generated strictly out of sample. Performance is relatively stable across tournaments, with RMSE values ranging between approximately 3 and 4.5 and R^2 consistently above 0.25, suggesting that the model generalizes reasonably well over time rather than being driven by a single favorable test split. Tournament 319 stands out with the strongest predictive performance, achieving an out-of-sample R^2 of 0.547 and the lowest RMSE, while Tournament 318 displays comparatively weaker results.

Table 4: Team-Level Win Prediction Accuracy

Metric	Value
Accuracy	0.736
Precision	0.777
Recall (Sensitivity)	0.743
Specificity	0.726
F1 score	0.759
Relative Gini	0.590

The team-level classification results in Table 4 demonstrate strong predictive performance when the outcome is simplified to a binary win indicator. The model achieves an accuracy of 0.736, with balanced precision and recall values, indicating that it performs well in identifying both winning and losing teams without excessive bias toward either class. The F1 score of 0.759 reflects a favorable trade-off between precision and sensitivity, while the relative Gini coefficient of 0.590 suggests substantial ranking power compared to a random classifier.

Table 5: Confusion Matrix

	Reference	
Prediction	**0**	**1**
0	77	35
1	29	101

The confusion matrix presented in Table 5 for the team-level win prediction reveals the robust classification performance of the XGBoost model. Out of 242 total observations, the model correctly identified 178 outcomes, consisting of 77 true negatives (predicted loss, actual loss) and 101 true positives (predicted win, actual win). The relatively low frequency of false negatives (29) and false positives (35) suggests that the model maintains a balanced error profile without significant bias toward over-predicting wins.

Now, we turn to the less successful part of our machine learning analysis: the predictive modeling of motion bias. Model evaluation based on RMSE, R-squared, and the gain curve indicates very poor predictive performance: the root mean squared error of 14.66 exceeds the standard deviation of the actual bias values (12.78), implying that the model performs worse than a naive benchmark that predicts the sample mean for all observations. This conclusion is reinforced by the negative R-squared value of -0.447, which shows that the model explains less variation than a horizontal line at the mean of the dependent variable. The gain curve likewise fails to demonstrate meaningful ranking or discrimination ability. Taken together, these results suggest that the model does not capture systematic patterns in bias and that further hyperparameter tuning is unlikely to yield substantive improvements.

4.3 Comparison of XGBoost and Standard Panel Data Methods

Table 6 compares the explanatory and predictive performance of econometric and machine learning approaches across different outcomes. For individual speaker performance, traditional econometric models achieve higher in-sample explanatory power, with pooled OLS yielding an R^2 of 0.586 and the fixed effects specification retaining a substantial within-speaker R^2 of 0.394. The machine learning regression model based on XGBoost attains a lower but still meaningful out-of-sample R^2 of

0.331. At the team level, the XGBoost classification model achieves an accuracy of 73.6%, demonstrating strong predictive performance for binary debate outcomes despite the limited success of linear econometric models in explaining aggregate team results.

Table 6: Econometric and Machine Learning Model Performance

Model	Method	Outcome	Performance metric
Pooled OLS	Econometric	Speaker points	$R^2 = 0.586$
Fixed effects	Econometric	Speaker points	$R^2 = 0.394$
XGBoost (regression)	Machine learning	Speaker points	$R^2 = 0.331$
XGBoost (classification)	Machine learning	Team win	Accuracy $= 0.736$

The summary in Table 7 highlights the central findings of the analysis regarding individual performance. Across econometric specifications, teammate quality consistently emerges as the dominant determinant of speaker scores, even after controlling for individual fixed effects. This result points to substantial peer or team effects and suggests that strong speakers tend to cluster within teams or institutions, reinforcing performance advantages. The absence of a significant gender effect across models further indicates that performance differences are not driven by gender composition. In parallel, the machine learning analysis confirms that individual performance is moderately predictable using historical and contextual features, although predictive accuracy remains well below a deterministic benchmark, reflecting the inherently noisy nature of debate evaluations.

Table 7: Summary of Main Findings

Question	Method	Key result
Individual performance	OLS / FE	Teammate quality dominates
Individual performance	XGBoost	Moderate predictive power
Motion bias	Text ML	No predictive signal
Team success	OLS	No compositional effects
Team success	XGBoost	73.6% accuracy

The remaining results summarized in Table 7 concern motion-level bias and team success. Text-based machine learning models applied to debate motions fail to uncover systematic predictive patterns in affirmative-side advantage, implying that motion wording alone does not reliably generate structural bias in outcomes. At the team level, simple econometric models show no significant effects of observable team composition, such as experience or gender mix, on ballots won. Nevertheless, when the outcome is modeled using a flexible classification approach, team success becomes substantially more predictable, with the XGBoost model achieving over 70% accuracy. Together, these findings suggest that while linear econometric models are well suited for identifying interpretable drivers of individual performance, richer machine learning methods are better equipped to capture complex, nonlinear patterns underlying team-level outcomes.